Bilkent University

Department of Computer Engineering

# Senior Design Project

## High Level Design Report

***Bil-Events : A mobile app to organize events in Bilkent University***

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High Level Design Report
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# 1. Introduction

According to YÖK's website, there are currently over 13.000 students studying in Bilkent [1]. Most of these students are the members of some student clubs and these clubs host many different events for Bilkenters to warm up to the college. However, there are several problems with the event notification system. First of all, since there is no unified notification system that people can access the information about events from various places. There are three main ways to access event information. One of them is from webmail notifications (BAIS), another one is from students clubs' posters and the third one is from SRS' GE 250/251 notifications. However, all of these three have their own flaws. To clarify what these flaws are actually, we should examine the event notification systems. The first one, the BAIS mail system, which sends over ten mail per day about not only events but also all notifications about Bilkent University and the seminars. It is hard to keep track of events and remember their dates. In addition to this, the BAIS mail system sends the emails unaesthetically, it causes resentment even for the students who want to invest their valuable time by attending the events. The second one, using paper posters is outdated, also a huge waste of paper and they are not well-distributed. They are so randomly distributed that it causes distraction among the students who want to find the right event for themselves. For instance, you can't find specific events posters in every building as they are not hung everywhere or they are overlapped when they are hung. For the students in the East Campus, this issue cannot even be mentioned because they cannot reach information about the events, because the students don't have any idea how to find the locations of the events and the events are generally held in the Main Campus, the distance between East Campus and the location of the events and lack of knowledge on how to find the location give resentment to the students who are in East Campus. BAIS e-mails and posters are not descriptive enough to handle that problem we mentioned. The last point we want to mention is that the update on SRS' event notification system does not even show the events that don't give GE 250/251 points. As a group of students who are in this system, we observed and suffered from these problems and as a solution we offer an app which will be accessed by everyone.

## 1.1 Purpose of the system

Our mobile app called Bil-Events which collects and organizes the activities better than the current system in Bilkent University. This app will be done with the purpose of easy access to events in Bilkent, advertising the activities more aesthetically, notifying Bilkenters in a better way as it is described below, creating the win-win situation between the clubs and the participants, increasing the competition between clubs, giving the opportunity to reach more participants for the clubs and pushing the club leaders and the members to be more active. Another expectation is to reduce paper consumption caused by posters. As we mentioned in above, Bil-Events is going to meet the students' needs. Bil-Events can efficiently and effectively be used by every student. In an efficient manner, the students can find the location and the starting time of the events more easily and in an organized way which the BAIS and the SRS lack of. The students will be notified by Bil-Events when the event is close to the starting time. In this way, the students can be prepared for the events more easily. In an effective manner, the students can eliminate the events which they don't want to attend and they can attend the similar events they previously attended. In addition to this, the students can see the pictures and the videos of the events to have an idea about how the event is happening. This situation will give the clubs the opportunity to present their activities better. The club members can contact enthusiastic participants to give the opportunity to work with them and create more impactful events. The collaboration may bring the competition indirectly, because organizing more impactful events can bring fame to the organizer club members. That may push other club members to be more active, be in touch with their followers etc. To summarize, we are creating the application which strengthens the relationships between the club members, the members and the students in Bilkent University.

## 1.2 Design goals

### 1.2.1. Usability

The application will be targeted at students and club leaders so the application's user interface will be simple and straightforward. Everyone with basic mobile knowledge should be able to use the app and the average user should become acquainted with all of the functionalities in less than 5 minutes. To this end, tutorials will be included in the app.

Since internet connection may not be accessible as it is possible, Bil-Events will work offline for the saved event' notification. For a more accurate diagnosis the users can refresh the events' information when they connect to the internet.

### 1.2.2. Performance

The response time for the finding events should be no more than 10 seconds if the internet connection is sufficient enough. The load time of the application should be no more than 5 seconds but it can change according to Android version and hardware capacity. The application should take no more than 2 second to switch between panels in order to offer a smooth user experience for our users.

### 1.2.3. Reliability

Our application will give importance to reliable information about the students in the university. The students which are from other universities or the students with the fake accounts cannot sign in to the Bil-Events. Every student should have only one account on the application and it should be the verified account by the BCC.

### 1.2.4. Maintainability

The system will be developed in a way to be flexible to possible changes in the future. The code will be well documented in order to understand and fix issues that may arise faster.

## 1.3 Overview

Bil-Events is a mobile application which offers essential services to club members and participants. This application makes events easily reachable to Bilkenters.The most important feature of Bil-Events is that all Bilkenters can access and participate in an event which is added by event leaders or MSSF(Müzik ve Sahne Sanatları Fakültesi). Events and users' information will be stored in our Database. Events will be categorized according to their tags and these events will be seen by the users who prefer these types of events thanks to the recommendation system. Also without recommendation, users can search and see all the events according to their date, hour and meeting place. Bil-Events will be created by using Android Studio and it will be available for only Android devices. As the interface will be not complex, users can easily use the app. It's interface design will be done by using Adobe Photoshop CC as we want all designs to belong to us.

# 2. Current software architecture

## 2.1 BAIS (Bilkent University Administrative Information System)

BAIS is a kind of notification way that should be changed for its unpractical services such as:

- Everything about University and student come as a BAIS mail to the students' email inbox that makes the inbox to be messy and hard to remember the events and their informations.
- If students want mails from Bilkent clubs, they have to accept mail terms on the SRS system for a specific club. Then the mail inbox become more messy than ever since all club notifications start to come as an email or multiple emails.

## 2.2 Social Media

Social media is a kind of notification way that should be changed for its unpractical services such as:

- Clubs' social media accounts can be found and followed but this problem is how students can know the existence of clubs.
- When events are cancelled, the notification on the social media may not be reached or notified before the event.

# 3. Proposed software architecture

## 3.1 Overview

To show the grouping of classes and functions to related systems and the interaction between different subsystems, The subsystem decomposition is used. In the diagrams, each subsystem and their I/O gets and produces to accomplish their part of work. They also interact with each other in order to make a connection. By having a well defined subsystem structure, our main purpose is to have less errors during the implementation and easily handle the changes when it is required. Considering the long term benefits and realizing the importance of this part, we spend a considerable amount of our time to come up with a well-defined subsystem structure.

## 3.2 Subsystem decomposition

Bil-Events will be built upon a database system that uses students preferences as an update matter. Client/Server architecture will be handled. Students can connect the app through the server. After every event is added to the system, the database will be changed accordingly. Same goes for students added to events as well. Client side will be focused on responding to user's output while the whole processing part will be handled by the server side.

We designed our overall system as 3 tier: User Interface Tier  which make up the Client side, and Processing and Filtering Tier and Data Tier which make up the server side. In the Server side Data tier stores the important data such as events, user information passwords etc. After the usage of the user interface the database will be updated.
The schema and the subsystems are explained clearly in the Subsystem Services section.

## 3.3 Hardware/software mapping

The users should connect to the Internet to use Bil-Events which gets data from database and sends data to database. Bil-Events uses a database to keep user profile information such as users' Bilkent ID, roles etc. , information of the events such as title, date, location, statistical data about the number of people that are interested in, going to or are registered to the upcoming events of the organization. Therefore, constant interaction with the database is an obligation for our application and it is the main reason why it requires an Internet connection.

The users' hardware will be their Android phones only. After connecting to the application via an Android phone, it will be connected to the application server. The server will be connected to the database server. We will use MySQL as the DBMS.

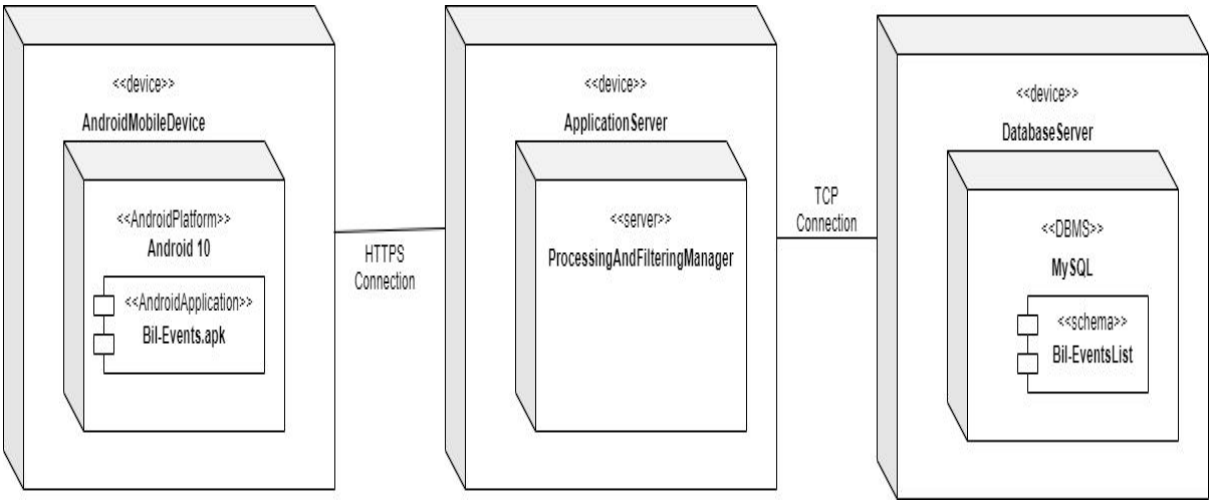
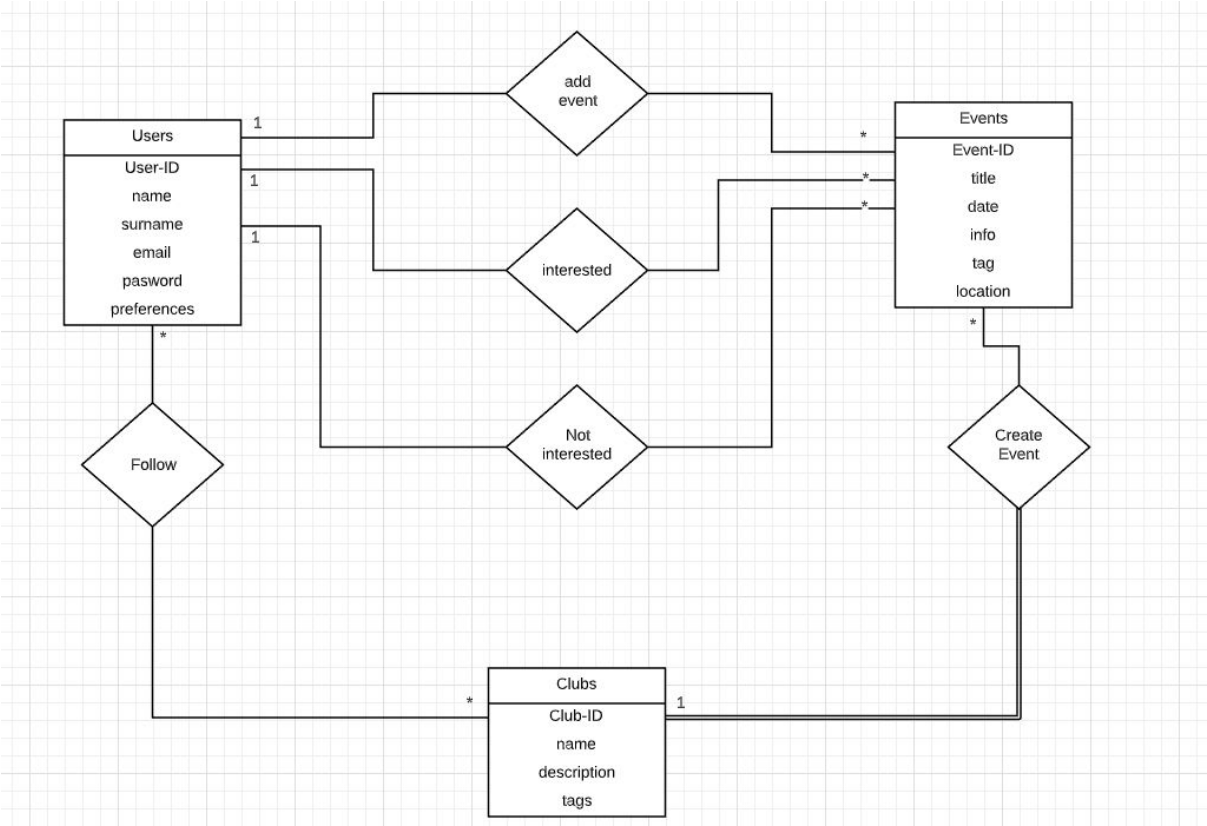



Figure 4. Hardware - Software Mapping

## 3.4 Persistent data management

All data related activities / information will be stored in the data we mentioned before.If the information of event changes it will be updated in the database as well. Such as the topic of the event, tags of the event is one of the main examples of event information. This applies to  user information as well. Password, birthdays, mail addresses etc. All user/organization accounts and their information such as user/organization name, email address can be changed or deleted if needed.




## 3.5 Access control and security

Bil-Events aims to bring all events notifications together in Bilkent University. This goal brings out some security openings that non-Bilkenters can access the events and users information can be accessed. So our users' security is of the topmost significance. Therefore, Bil-Event offers various security measures such as credential security and event security. To add more, the design and implementation without bug will be carried out. So the system will be monitored and updated constantly to prevent any inconsistencies, conflicts and outside accesses.

### 3.5.1 Credential Security

- All Bilkenters must provide an ID that belongs to Bilkent University and a password to register into the system. In order to keep passwords as secure as possible, Bil-Events applies a password policy of at least 6 characters that contains at least 1 upper-case letter and 1 numerical digit just for strong password requirements. Furthermore, all passwords are encrypted with triple DES encryption algorithm which

stands for “Triple Data Encryption”. It is a symmetric-key block cipher that applies a well-known DES algorithm three times to each data block [1].

- Only one account with a unique Bilkent University ID can be associated with one Bilkent University e-mail address. Therefore, multiple clone accounts will be prevented.
- For admins which are the club leaders or MSSF, authorization must be required in order to register them into the system. Admins must provide their credentials to protect clubs from fake club accounts which is not possible since clubs are checked their existence in Bilkent University.
- Bil-Events keeps all credential information on its database. For this reason, a secure database design and implementation which must be reliable will be carried out. In order to prevent database attacks such as SQL Injection to hack our database, parameterized queries will be used [2].
- No information about users will be published and shared.

### 3.5.2 Event Security

Our platform can only be reliable if admins provide relevant events. Therefore, Bil-Events checks every event starting from its creation to its end. High-user satisfaction and safety are platforms’ main concerns. Hence, Bil-Events provides below features to meet those targets:

- Bil-Events checks every events to prevent any undesirable consequences like fake events, illegal events, discrimination and etc. during its creation. This check is done by evaluating creator’s user-ratings and comments. Also, a complaint option is offered for any users to notify us about their concerns about a specific event or the club. We will evaluate every complaint to decide on the events’ status.
- All admins will be punished immediately if they involve any kind of illegal and forbidden acts. To exemplify, fake or unreliable events can be added by club leaders. In this situation, their club rating will be reduced by the system and might be disinterested from other users for further events. So club leaders should be careful while adding events.
- After an event finishes, participants can evaluate and comment the event also can help the club to make progress. We do not want dishonest evaluations to harm clubs so unreliable comments and evaluations will be checked. If considered reasonable,

evaluation or comment owners will be banned from adding comment and evaluating clubs.

## 3.6 Global software control

A centralized event-driven control system is driven by Bil-Events. Bilkenters will login into the system by using their university IDs and passwords. If the ID is not matched on database, the user is supposed to be not a Bilkenter so logging in will be unsuccessful. Users' are given different feature accesses according to their types like admin as club leaders or students. While the events are distinguished by using a tag on the database, users can continue to log in simultaneously. Therefore multiple users can request services at the same time from the system. To handle service errors, all request servicing mechanisms are designed to service various users concurrently. To clarify, user requests are queued first then the requests serviced one by one by Bil-Event's server to prevent any conflicts and overloads. Moreover, any updates on the database are reflected to relevant tables in atomic time without servicing any further requests by using trigger based transaction management. The purpose of this is to secure the event platform and to provide a nonstop environment. On the other hand, the event-driven nature of the system implies that the flow of the program is determined by user actions. System is designed to service user requests when there is an available one and stays idle when there is no action.

## 3.7 Boundary Conditions

Bil-Events is going to include various boundary conditions. These boundary conditions can be stated as initialization, termination and failure.

### 3.7.1 Initialization

Users can access the application on an Android phone. The application requests user for user credentials, the validity of the user is checked (e.g. whether the user is a member of Bilkent University), if the user is valid then the user can sign up. Users will have to log in using their Bilkent ID and password. If user types his/her credentials wrong, sign up page will be displayed again.

### 3.7.2 Termination

When the connection between the client and the server is ended, the system is terminated.

The user will be able to logout of the system with the logout button located at the profile page. When the minimum requirements of sustainability of the application cannot be met (e.g. the battery charge is over); therefore, the user is logged off as well.

### 3.7.3 Failure

If the application is terminated while it is performing an action, this situation can lead to some failures. The user can encounter some failures during the update of the application.

# 4. Subsystem services

Current designs are not the exact solutions to the problem, because they have a hard time presenting the information to the students. The current systems are not user friendly, it is not organized and it is presented with the other emails or irrelevant information. Our application is portable and presentable to get the information of the events efficiently. Combining the portability, the specialization of event representations and the recommendation system will make our platform preferable. Below we will explain our two subsystem design in a more detailed way.

## 4.1. Client Subsystem

Our client subsystem is composed of two tiers of labels: User Interface Tier and Control Tier.
The UI Tier interacts with the end user and controls the interactions in order to respond accurately to the end users call and to communicate with the server, it has the Control Tier. Our application will be an Android application, we have designed the client subsystem accordingly. Client Subsystem will be responsible for all of the user operations, sending and receiving data, notifications, displaying information about the events etc. End users can make different requests and our Client Subsystem can handle accordingly.

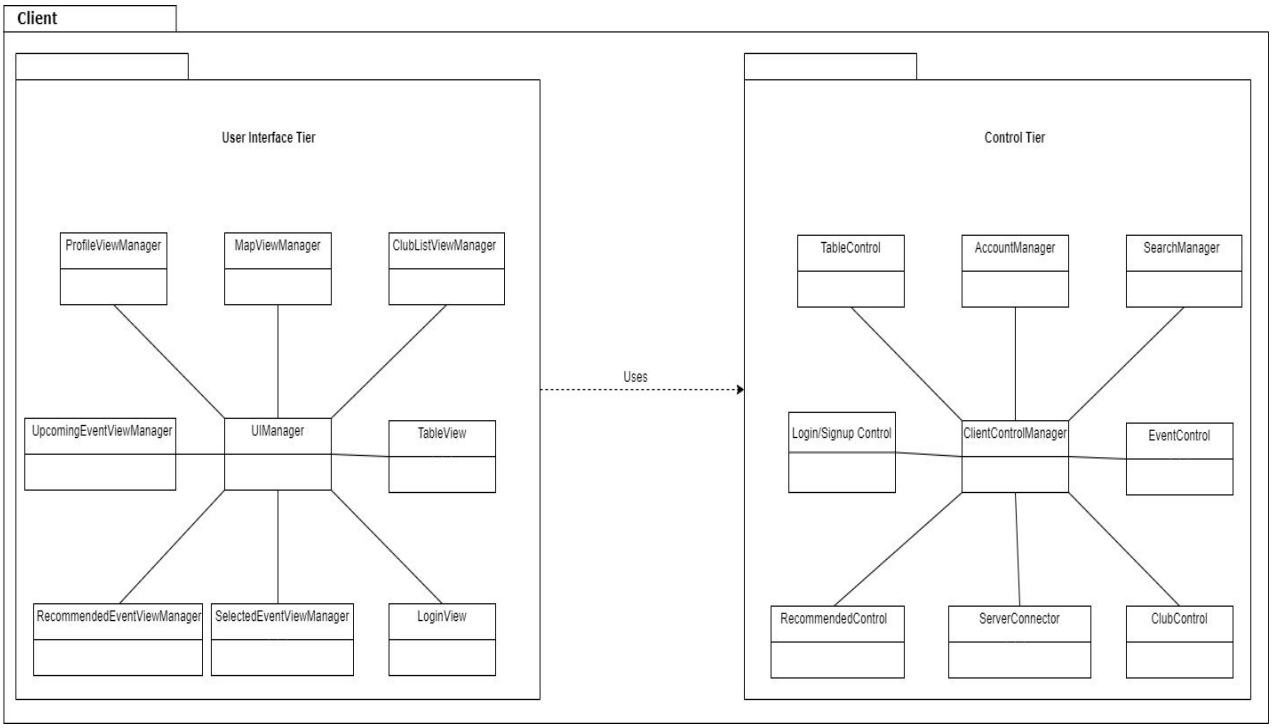



Figure 6. Client Subsystem

### 4.1.1. User Interface Tier

Inside the UI Tier, the application handles all of the user input operations. The tier consists of the following classes.

**ProfileViewManager:** The class handles the request of viewing the personal profile of the user of the application.
**UpcomingEventViewManager:** The class handles the request of viewing the upcoming events in the application.
**MapViewManager:** The class handles the request of viewing the location of the events in the application.
**RecommendedEventViewManager:** The class handles the request of viewing the recommended events in the application.
**SelectedEventViewManager:** The class handles the request of viewing the selected events in the application.
**LoginView:** The class handles the request of viewing the login page of the application.
**TableView:** The class handles the request of viewing the main menu page of the application.

**ClubListViewManager:** The class handles the request of viewing the list of all available clubs in the application.

### 4.1.2. Control Tier

The Control Tier handles all of the saving and receiving data operations and it is directly connected to the server. The tier consists of the following classes.

**TableControl:** The class handles the control of actions on the main menu in the application.
**AccountManager:** The class has the information and the verification of the account of a user in the application.
**SearchManager:** The class handles the request of searching for any event in the application.
**Login/Signup Control:** The class handles the request of login/signup operations in the application.
**RecommendedControl:** The class handles the request of finding the best possible recommended events in the application.
**ClubControl:** The class has the information and the verification of the clubs in the application.
**EventControl:** The class has the information and the verification of the events of the clubs in the application.
**ServerConnector :** The class creates the connection between the client and the server.

## 4.2. Server Subsystem

The Server subsystem is composed of two tiers: Processing and Filtering Tier and Data Tier. Processing and Filtering Tier is responsible for getting all the requests, processing them and filtering all the data as well. Data Tier is responsible for storing the data. The information of new users, new events or new clubs are sent by the client and after the processing part the data is sent to the server. Because, some of the requests may contain irrelevant information that will be processed and checked by this tier. For instance, the students from other universities can access the application and the students must be checked whether the students are from Bilkent University or not. Another example could be the addition of non-existent clubs to the application or forgetting the deletion of inactive clubs; therefore, such small details like the examples above need to be checked.

After the addition of the information of any event, any club or any student is successfully done, the students can access the events in detail, they can see the upcoming events, the selected events and the recommended events as well . For accessing the event, the server part is responsible since the client is requesting. Server side by using the secure database implementation can detect phishing attempts of any violent user. In this way, the server subsystem controls and secures the data and executes all the operations with them.

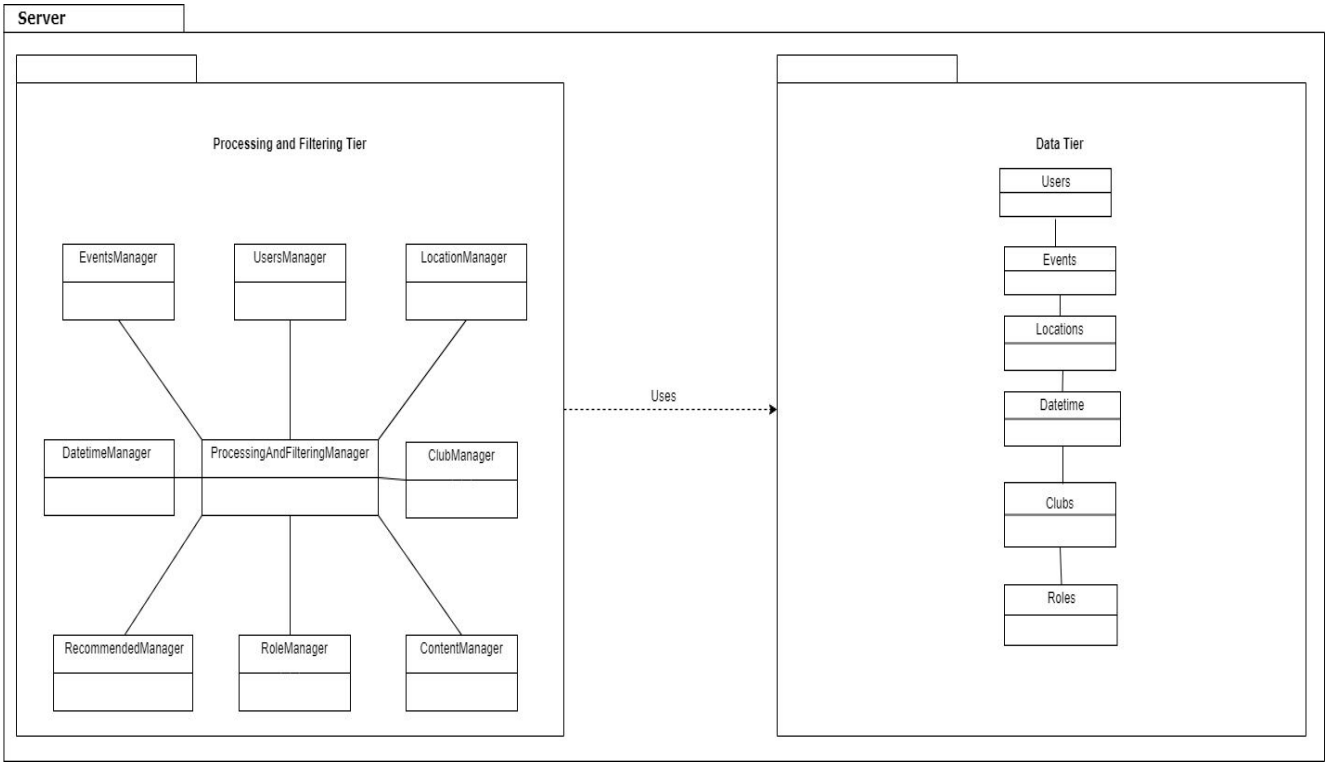



Figure 7. Server Subsystem

## 4.2.1. Processing and Filtering Tier

Processing and Filtering Tier is responsible for processing and filtering all of data before saving them in the database or making a prediction to generate data using existing data segments. The classes of Processing and Filtering Tier are explained in below.

**EventsManager:** The class checks the validity of the events of the application.
**UsersManager:** The class checks the validity of the users (e.g. whether the user is a member in Bilkent University) in the application.
**LocationManager:** The class checks the validity of the locations in the Bilkent University, in the application.

**ClubManager:** The class checks the validity of the clubs in the Bilkent University, in the application.
**ContentManager:** The class checks the validity of the content of an event in the Bilkent University, in the application.
**RoleManager:** The class checks the validity of the role of the member in the Bilkent University, in the application.
**RecommendedManager:** The class makes the prediction of the events that the user may possibly prefer to attend by using the previous attended events of the user.
**DatetimeManager:** The class checks the validity of the starting date and time of the events in the Bilkent University, in the application.

### 4.2.2. Data Tier

Data Tier stores all of the raw data by organizing them and handles all of the queries when the data is needed to be accessed. The tier consists of the following classes.

**Users:** The class stores the users' information in the application.
**Events:** The class stores the events' information in the application.
**Locations:** The class stores the information of the locations of Bilkent University in the application.
**Datetime:** The class stores the start date and time of the events in the application.
**Clubs:** The class stores the clubs' information in the application.
**Roles:** The class stores the information of the roles of the members of Bilkent University in the application.

# 5. New Knowledge Acquired and Learning Strategies Used

New knowledge that we have learned while we are designing the project is the main goal of this project. For this purpose, we'll use an online learning strategy which we already discussed and found online courses. For the others, we will use web search methods, articles and analyze other applications. The new knowledges that we are learning to make an efficient, trustworthy and reliable application are given in the list below:

- Android Studio

- Android programming
- Recommendation Engine
- DES algorithm for encrypting
- Database encryption
- Server handling
- Photoshop CC for design purposes

## 5.1 Android Studio

We found an Android Studio course from Udemy in order to learn both the basic and advanced structure of Android programming[3]. For now, we are practicing while watching the course. After having finished the course, we will start actual implementation of Bil-Events.

## 5.2 Recommendation Engine

Also the recommendation engine is the one of crucial part of our project since we don't want Bil-Events to recommend users irrelevant events that users won't interest. Therefore, we will learn all steps of the recommendation engine such as collaborative filtering and its types, data needed to build a recommender, libraries available to build recommenders, use cases and challenges of collaborative filtering.

## 5.3 DES Algorithm Encryption

To add more, DES algorithm will be used for encrypting[1]. To learn DES algorithms, the first action we do is that we aim to understand how it works. Then we will implement the algorithm for our project. After the implementation, testing the system and correcting our mistakes, if any, are our aim.

## 5.4 Database Encryption

Database encryption is to protect the data stored in a database from being accessed by individuals with potentially "malicious" intentions. There are lots of methods for encryption so the most efficient one with high performance will be used. MySQL will be used for databases.

## 5.5 Server Handling

Creating server and server handling are concepts that we are non familiar with. Users need a server which can handle a user's request from the phone. Basically users need a web-service API with a connected database. Therefore, we'll learn SSL or TLS which is a common building block for encrypted communications between clients and servers[4].

## 5.6 Adobe Photoshop CC

Adobe Photoshop CC will be used for design purposes. We already used it for our logo and mock ups. For further background and images that are required in the application will be done by using Photoshop CC.

# 6. Glossary

**BAIS :** Bilkent University Administrative Information System
**DES :** Data Encryption Standard
**SSL :** The Secure Sockets Layer
**TLS :** Transport Layer Security

# 7. References

[1] Seung-Jo Han, Heang-Soo Oh and Jongan Park, "The improved data encryption standard (DES) algorithm," *Proceedings of ISSSTA'95 International Symposium on Spread Spectrum Techniques and Applications*, Mainz, Germany, 1996, pp. 1310-1314 vol.3, doi: 10.1109/ISSSTA.1996.563518.

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